

WASP-50b

R-1268

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_201106 · created 2026-07-31T13:32:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459159.936 +/- 0.04 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.051
Transit depth (Rp/Rs)^2	2.85 +/- 1.72 [%]
Orbital Inclination (inc)	86.45 +/- 2.88
Airmass coefficient 1 (a1)	1.12 +/- 0.65
Airmass coefficient 2 (a2)	-0.027 +/- 0.083
Scatter in the residuals of the lightcurve fit is	57.42 %
Best Comparison Star	None
Optimal Aperture	1.23
Optimal Annulus	9.46
Transit Duration (day)	0.063 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.051 vs 0.1404 (deviation 0.35 σ)
Duration fit vs published	0.063 d vs 0.0754 d (deviation 0.27 σ)
Mid-time O–C	60.8 min
Depth SNR	2.0
Residual scatter	57.42 %

Light curve

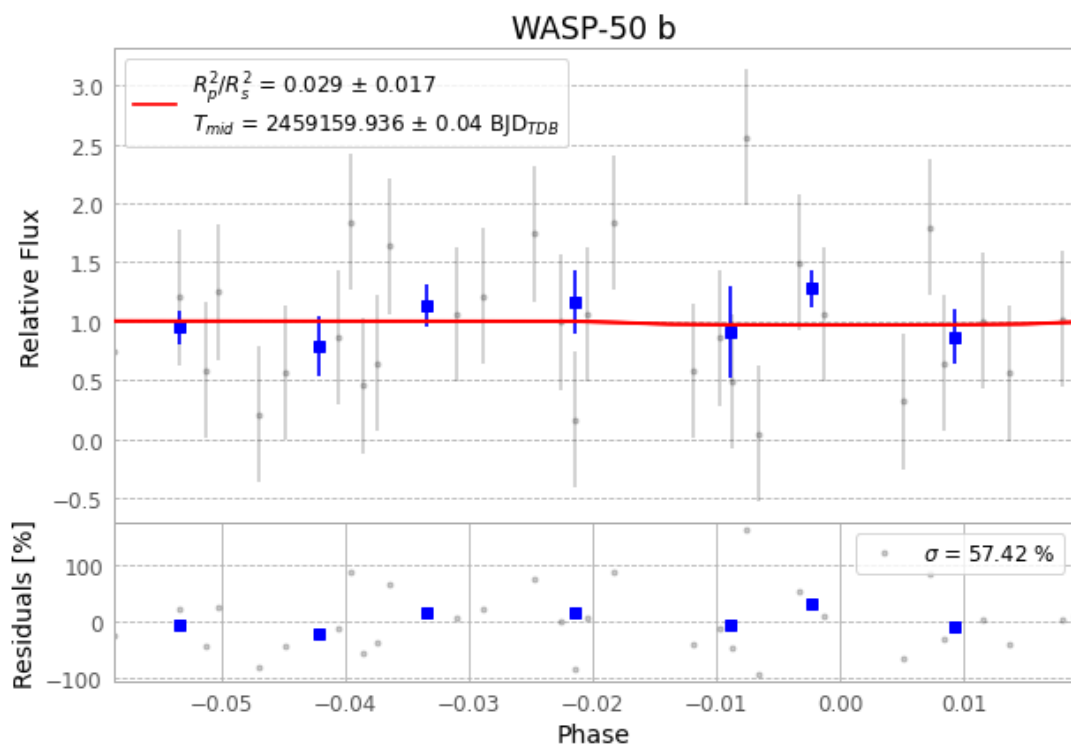


Figure 1 — FinalLightCurve_WASP-50 b_2020-11-06.png (SHA-256 44a2e9b078689166...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [535, 148], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2bb3a32baecd8c4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50201106072712.FITS → WASP-50201106111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-201106.log (0c6f4579a3d1...), FinalLightCurve_WASP-50 b_2020-11-06.png (44a2e9b07868...), FinalLightCurve_WASP-50 b_2020-11-06.csv (0338e54d8780...)

Compute resources

Wall time 250.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 13:32Z.